Milad Tatar Mamaghani | Ph.D.

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LinkedIn ▪ Homepage ▪ Google Scholar ▪ ORCiD ▪ GitHub

Education

Doctor of Philosophy (Ph.D.) in Engineering — **Melbourne, Australia**
Monash University - Clayton Campus — *Jan. 2019–Dec. 2022*
Thesis: *Safeguarding Beyond-5G Wireless UAV Communications: Design and Optimization*

Bachelor of Science (B.Sc.) in Electrical–Communications Engineering — **Tehran, Iran**
Amirkabir University of Technology (Tehran Polytechnic) — *Sep. 2012–Oct. 2016*
Thesis: *Secure Communications with Untrusted Relaying and Wireless Energy Harvesting*
Also awarded B.Sc. in Control Engineering with additional course credits (Feb. 2018)

Research & Professional Experience

Postdoctoral Research Fellow (Level B) — **Canberra, Australia**
The Australian National University — School of Engineering — *Jan. 2023–Jan. 2025*
- ✓ Conducted research on physical-layer security, short-packet communications, and integrated sensing and communications.
- ✓ Published in leading venues, contributed to award-winning work, supervised students, and supported research activities.
- ✓ Mentor: A/Prof. Xiangyun Zhou

Independent Consultant — **Sydney, Australia**
Mercor / Scale AI — Remote — *Feb. 2025–Present*
- ✓ Contribute to research-oriented AI projects involving technical evaluation, model reasoning, and domain-specific analysis.
- ✓ Maintain an independent research profile in wireless communications and AI-enabled network design.

Doctoral Researcher — **Melbourne, Australia**
Monash University — Department of Electrical and Computer Systems Engineering — *Jan. 2019–Dec. 2022*
- ✓ Developed optimization and machine-learning/AI methods for energy-efficient secure wireless aerial networks.
- ✓ Researched UAV, MIMO, beamforming, THz, covert communications, and intelligent reflecting surface systems.
- ✓ Supervisor: A/Prof. Yi Hong

Research Assistant — **Tehran, Iran**
Amirkabir University of Technology — Wireless Communication Research Lab — *Jan. 2016–Nov. 2018*
- ✓ Researched physical-layer security, untrusted relay networks, wireless energy harvesting, and cooperative jamming.
- ✓ Supervisor: Prof. Abbas Mohammadi

Research Intern — **Tehran, Iran**
Iran Telecommunication Research Center (ITRC) — Fixed Communications Group — *Jun. 2016–Sep. 2016*
- ✓ Completed a summer research internship on applied telecommunications systems and national network infrastructure.

Research Interests

- AI-Native Wireless 6G Networks
- Machine Learning/AI for Communications
- UAV, Satellite, and Non-Terrestrial Networks
- Integrated Sensing and Communications
- Physical-Layer Cybersecurity and Privacy
- Statistical Signal Processing
- MTC/URLLC/RIS/THz
- Deep Reinforcement Learning

Honours & Recognitions

- **Best Paper Award, IEEE SPCC-TC (2024)** [Certification]
  - ✓ Recognises outstanding contributions to Signal Processing and Computing for Communications in IEEE ComSoc Community – only one paper is selected each year.

- **Best Paper Award, IEEE ICC (2024)** [Certification]
  - ✓ Our work was selected as top 0.6% of 2,364 submissions across all symposia.
- Australian **Global Talent** Recognition — National Innovation program, 2024
  - ✓ Recognises individuals of exceptional achievement in strategically important fields.
- **Exceptional Ph.D. Thesis** Evaluation, Monash University, 2022
- **Vice Chancellor's Congratulatory Letter** for Teaching and Learning Excellence, Monash University, 2019
  - ✓ Awarded for outstanding performance in teaching and learning based on student unit evaluation.
- **Exceptional Talent** Recognition, Amirkabir University of Technology, 2014–2016

Awards & Grants

- **Early Career Researcher International Travel Grant** (AUD \$3000), Australian National University, 2023
- **Postgraduate Publications Award** (AUD \$5000), Monash University, 2022
  - ✓ Competitive merit-based award for outstanding doctoral publications.
- **Graduate Research Completion Award**, Monash University, 2022
- **Engineering Faculty's Postgraduate Research Scholarship**, Monash University, 2022
- **Monash Graduate Scholarship + International Tuition Scholarship**, Monash University, 2019–2022
  - ✓ Full stipend (AUD $120,000) and tuition waiver (AUD $200,000); competitive merit-based selection.

Teaching Experience

Teaching Associate — **Melbourne, Australia**
Monash University — Faculty of Engineering — *Feb. 2019–Dec. 2022*

✓ Facilitated hands-on learning through effective tutoring, practical sessions, and lab demonstrations.
✓ Provided mentorship that enhanced student performance in coursework and engineering design projects.
✓ Contributed to curriculum delivery by designing assessments and offering constructive feedback to students.
✓ Coordinated and supervised engineering design projects for undergraduate and postgraduate students.
✓ Mentored students, assisted in designing assessments, marking, and evaluating presentations.
✓ Achieved an overall student satisfaction score of 4.73/5.0, ranking in the top 7.3% of all University units in the semester.

Courses Taught

- ECE5179: Neural Networks and Deep Learning; Tutor and Lab Demonstrator, Semester 2, 2022
- ECE5883: Advanced Signal Processing; Lab Designer and Demonstrator, Semester 1, 2022
- ENG5001: Advanced Engineering Data Analysis; Tutor, Semester 1, 2022
- FIT1048: Fundamentals of C++; Tutor and Lab Demonstrator, Semester 1, 2022
- ECE5884: Wireless Communications; Lab Demonstrator, Semester 2, 2021
- ECE4191: Engineering Design; Lab Demonstrator and Supervisor, Semester 2, 2020–2022
- ENG5105: Integrated Design; Coordinator and Project Mentor, Semester 2, 2020–2022
- ECE4146: Multimedia Technologies; Lab Demonstrator, Semester 2, 2020–2022
- ECE3141: Information and Networks; Lab Demonstrator, Semester 1, 2020–2022
- ECE4044: Telecommunications Protocols; Lab Demonstrator, Semester 1, 2020
- ECE2072: Digital Systems; Tutor and Lab Demonstrator, Semester 1, 2020–2022
- ECE4042: Communications Theory; Tutor and Lab Demonstrator, Semester 1, 2019–2022
- ECE4076: Computer Vision; Tutor, Lab Demonstrator, and Exam Coordinator, Semester 1, 2019–2022
- ECE2071: Computer Organisation and Programming, Tutor and Lab Demonstrator, Semester 1, 2019–2022

eAssessment Supervisor — **Melbourne, Australia**
Monash University — eSolutions & Student Services — *Jan. 2020–Dec. 2022*

✓ Supervised exams and supported integrity, compliance, and student-facing assessment operations.

Teaching Assistant — **Tehran, Iran**
Amirkabir University of Technology — Department of Electrical Engineering — *Jan. 2016–Oct. 2017*

✓ Taught undergraduate courses in communications systems and computer programming as a lab instructor.

Student Supervision

- **Yu Li**, Ph.D. Candidate, ANU, 2023–Present
  - ✓ Topic: Physical Layer Anonymous Communications; Co-supervised with A/Prof. Xiangyun Zhou
  - ✓ Contributed a co-authored paper (Ref. [18]) submitted to IEEE Trans. Commun. (2026).
- **Mac Gilligan**, Engineering R&D Student, ANU, 2024–2025
  - ✓ Topic: SDR Implementation for Over-the-air Secure File Transmission with GNU Radio
  - ✓ Nominated for ANU's University Medal for outstanding academic record
- **Reetika**, Visiting Bachelor Student, Indian Institute of Space Science and Technology, 2023–2024
  - ✓ Topic: Physical layer security for Terahertz communications; Co-supervised with Prof. Nan Yang
- **Team Supervision**, Engineering Design Projects, Monash University, 2021–2022
  - ✓ Robotic Arm: Automated color cube sorting robotic arm with microprocessor controller, 2021
  - ✓ Mobile Robot: Smart mobile robot with odometric sensing for high-level navigation and ball collection, 2022
  - ✓ Baby Monitor: Sustainable baby monitor with voice and image recognition and parent alarm system, 2022

Professional Service & Leadership

Conference Organisation

- **Technical Program Committee Member** — IEEE Conference on Vehicular Technology, 2024
- **Session Chair** — IEEE Global Communications Conference (Symposium on Physical Layer Security), 2023
- **Selection Committee Panel Member**, ANU Summer Research Scholarship Programme, 2023

Peer Review

- **Technical Reviewer** for over 25 international journals and conferences with $140+$ reviews, 2018–2025
  - **Journals:** IEEE J. Sel. Areas Commun. (JSAC), IEEE Trans. Wireless Commun. (TWC), IEEE Trans. Commun. (TCOM), IEEE Trans. Inf. Forensics Security (TIFS), IEEE Trans. Signal Process. (TSP), IEEE Trans. Veh. Technol. (TVT), IEEE Wireless Commun. Lett. (WCL), and $10+$ additional journals.
  - **Conferences:** IEEE ICC (2020–2023), IEEE GLOBECOM (2017, 2024), IEEE VTC (2024), etc.

Memberships

- Member of IEEE (2022–2026)
- Graduate Student Member of IEEE (2020–2022)
- Member of IEEE Consultants Network (2020–2025)
- IEEE Communications Society (2020–2025)
- IEEE Signal Processing Society (2020–2022)
- IEEE Young Professionals (2020–2025)

Selected Talks & Presentations

[5] On the Information Leakage Performance of Secure Finite Blocklength Transmissions over Rayleigh Fading Channels, presented at the *Communication Theory Symposium: Physical Layer Security and Privacy* session, IEEE ICC 2024, Denver, CO, USA, June 2024. [Slides][Certificate]

[4] On the Secrecy Performance of Finite Blocklength Communications over Fading Channels, presented at the *21st Australian Communications Theory Workshop (AusCTW)*, Melbourne, Feb. 2024. [Poster]

[3] Secure Short-Packet Transmission with Aerial Relaying: Blocklength and Trajectory Co-Design, presented at the *Main Symposium on Communication and Information System Security (CISS): Physical Layer Security II* session at IEEE Globecom 2023, Kuala Lumpur, Malaysia, Dec. 2023. [Slides][Certificate]

[2] Final Review Seminar (Ph.D. Thesis Defense): Safeguarding Wireless Communications with Unmanned Aerial Vehicles: Design and Optimization, presented at the ECSE Department, Monash University, Melbourne, Australia, Feb. 2022. [Slides]

[1] B.Sc. Thesis Defense: Secure Communications via Untrusted Relaying with Wireless Energy Harvesting: Performance Analysis and Trade-offs [in Persian], presented at the Department of Electrical Engineering, Amirkabir University of Technology (Tehran Polytechnic), Tehran, Iran, Sep. 2016.

Publications

Scholarly outputs: 12 journal articles, 3 conference papers, 2 theses, and 2 manuscripts (in progress)

Research Impact: 650+ citations, h-index 11, i10-index 12 (as of May 2026)

[19] **Milad Tatar Mamaghani**, Xiangyun Zhou, Nan Yang, and Lee Swindlehurst, "Securing integrated sensing and communication against a mobile adversary: A Stackelberg game with deep reinforcement learning," ***IEEE Journal on Selected Areas in Communications***, vol. 44, pp. 942-958, Sep. 2025. (IF 17.2, **Q**1)
10.1109/JSAC.2025.3611404 | **PDF** | **arXiv** | **Code**

[18] Yu Li, **Milad Tatar Mamaghani**, Xiangyun Zhou, and Nan Yang, "Symbol-Aware Precoder Design for Physical-Layer Anonymous Communications," submitted to *IEEE Transactions on Communications*, Mar. 2026.
10.48550/arXiv.2602.20874 | **arXiv**

[17] Azadeh Tabeshnezhad, **Milad Tatar Mamaghani**, Lee Swindlehurst, Tommy Svensson, and Erik Ström, "Robust Beamforming Design for Secure Uplink NOMA-ISAC," under preparation, May 2026.

[16] **Milad Tatar Mamaghani**, Xiangyun Zhou, Nan Yang, Lee Swindlehurst, and Vincent Poor, "Performance analysis of finite blocklength transmissions over wiretap fading channels: An average information leakage perspective," ***IEEE Transactions on Wireless Communications***, May 2024. (IF 10.4, **Q**1)
10.1109/TWC.2024.3400601 | **PDF** | **arXiv** | **Code**

[15] **Milad Tatar Mamaghani**, Xiangyun Zhou, Nan Yang, Lee Swindlehurst, and Vincent Poor, "On the average information leakage of finite blocklength transmissions over Rayleigh fading channels," *in Proc.* ***IEEE International Conference on Communications***, Denver, CO, USA, pp. 1467-1472, June 2024.
10.1109/ICC51166.2024.10622223 | **arXiv**

[14] **Milad Tatar Mamaghani**, Xiangyun Zhou, Nan Yang, and Lee Swindlehurst, "Secure short-packet communications via UAV-enabled mobile relaying: Joint resource optimization and 3D trajectory design," ***IEEE Transactions on Wireless Communications***, Dec. 2023. (IF 10.4, **Q**1)
10.1109/TWC.2023.3344802 | **PDF** | **arXiv** | **Code**

[13] **Milad Tatar Mamaghani**, Xiangyun Zhou, Nan Yang, and Lee Swindlehurst, "Secure short-packet transmission with aerial relaying: Blocklength and trajectory co-design," *in Proc.* ***IEEE Global Communications Conference***, Kuala Lumpur, Malaysia, pp. 5998-6004, Dec. 2023.
10.1109/GLOBECOM54140.2023.10437972 | **arXiv**

[12] **Milad Tatar Mamaghani**, "Safeguarding Beyond-5G Wireless Communications with Unmanned Aerial Vehicles: Design and Optimization," Ph.D. Thesis, Electrical and Computer Systems Engineering Department, Faculty of Engineering, Monash University, Melbourne, Australia, Dec. 2022.
10.26180/21721709.v1

[11] **Milad Tatar Mamaghani** and Yi Hong, "Aerial intelligent reflecting surface-enabled terahertz covert communications in beyond-5G Internet of Things," ***IEEE Internet Things Journal***, vol. 9, no. 19, pp. 19012-19033, March 2022. (IF 9.9, **Q**1)
10.1109/JIOT.2022.3163396 | **PDF** | **arXiv** | **Code**

[10] **Milad Tatar Mamaghani** and Yi Hong, "Terahertz meets untrusted UAV-relaying: Minimum secrecy energy efficiency maximization via trajectory and communication co-design," ***IEEE Transactions on Vehicular Technology***, vol. 71, no. 5, pp. 4991-5006, Feb. 2022. (IF 6.41, **Q**1)
10.1109/TVT.2022.3150011 | **PDF** | **arXiv**

[9] **Milad Tatar Mamaghani** and Yi Hong, "Joint trajectory and power allocation design for secure artificial noise aided UAV communications," ***IEEE Transactions on Vehicular Technology***, vol. 70, no. 3, pp. 2850-2855, Mar. 2021. (IF 6.41, **Q**1)
10.1109/TVT.2021.3057397 | **PDF** | **arXiv**

[8] **Milad Tatar Mamaghani** and Yi Hong, "Intelligent trajectory design for secure full-duplex MIMO-UAV relaying against active eavesdroppers: A model-free reinforcement learning approach," ***IEEE Access***, vol. 9, pp. 4447-4465, Dec. 2020.
10.1109/ACCESS.2020.3048021 | **PDF**

[7] **Milad Tatar Mamaghani** and Yi Hong, "Improving PHY-security of UAV-enabled transmission with wireless

energy harvesting: Robust trajectory design and communications resource allocation," ***IEEE Transactions on Vehicular Technology***, vol. 69, no. 8, pp. 8586-8600, May 2020. (IF 6.41, **Q**1)
10.1109/TVT.2020.2998060 | **PDF** | **arXiv**

[6] **Milad Tatar Mamaghani**, Ali Kuhestani, Hamid Behrouzi, "Can a multi-hop link relying on untrusted amplify-and-forward relays render security?," ***Wireless Networks***, vol. 27, pp. 795–807, Nov. 2020.
10.1007/s11276-020-02487-w | **PDF** | **arXiv**

[5] **Milad Tatar Mamaghani** and Yi Hong, "On the performance of low-altitude UAV-enabled secure AF relaying with cooperative jamming and SWIPT," ***IEEE Access***, vol. 7, pp. 153060-153073, Oct. 2019.
10.1109/ACCESS.2019.2948384 | **PDF** | **Code**

[4] **Milad Tatar Mamaghani** and Robert Abbas, "Security and reliability performance analysis for two-way wireless energy harvesting based untrusted relaying with cooperative jamming," ***IET Communications***, vol. 13, no. 4, pp. 449-459, Mar. 2019.
10.1049/iet-com.2018.5718 | **PDF**

[3] **Milad Tatar Mamaghani**, Ali Kuhestani, and Kai Kit Wong, "Secure two-way transmission via wireless-powered untrusted relay and external jammer," ***IEEE Transactions on Vehicular Technology***, vol. 67, no. 9, pp. 8451-8465, July 2018. (IF 6.41, **Q**1)
10.1109/TVT.2018.2848648 | **PDF** | **arXiv**

[2] **Milad Tatar Mamaghani**, Abbas Mohammadi, Phil Yeoh, and Ali Kuhestani, "Secure two-way communication via a wireless-powered untrusted relay and friendly jammer," *in Proc.* ***IEEE Global Communications Conference***, Singapore, pp. 1-6, Dec. 2017.
10.1109/GLOCOM.2017.8253999 | **arXiv**

[1] **Milad Tatar Mamaghani**, "Secure Communications via Untrusted Relaying with Wireless Energy Harvesting: Performance Analysis and Trade-offs," *B.Sc. Thesis* [in Persian], Department of Electrical Engineering, Amirkabir University of Technology (Tehran Polytechnic), Tehran, Iran, Sep. 2016.

Key Collaborators

- Xiangyun Zhou — Associate Professor, ANU.
- Lee Swindlehurst — Distinguished Professor and Department Chair, UC Irvine.
- Nan Yang — Professor & Research Cluster Lead, ANU.
- Yi Hong — Associate Professor, Monash University.
- Vincent Poor — Michael Henry Strater Professor, Princeton University.
- Abbas Mohammadi — Professor, Amirkabir University of Technology.
- Kai Kit Wong — Professor and Chair in Wireless Communications, UCL.
- Ali Kuhestani — Assistant Professor, Qom University of Technology.
- Phil Yeoh — Senior Lecturer, University of the Sunshine Coast.

Languages

English (Fluent) | Persian/Farsi (Native) | Azerbaijani (Native)

Last updated: May 2026